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Shock Absorbing Helmet Liner Device

Abstract

A shock absorbing helmet liner device for improving shock absorption and inhibiting head injuries includes a liner coupled to an internal surface of a helmet shell. The liner absorbs a force of an impact to the helmet shell to inhibit the force of the impact from injuring a user who is wearing the helmet shell. The liner has a first layer that includes a multi-density polymer cushioning material. A second layer is coupled to the first layer. The second layer includes an air bladder having an interior space that is filled with a gas. A third layer is coupled to the second layer. The third layer includes a viscoelastic rubber material that absorbs the force of the impact and disperses energy from the force of the impact across the third layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

[0003] Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

[0004] Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

[0005] Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

[0006] The disclosure relates to helmets and more particularly pertains to a new helmet for improving shock absorption and inhibiting head injuries.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

[0007] The prior art relates to helmets. Helmets are vital for preventing head injuries or reducing the severity of head injuries. Helmets are therefore used in many potentially dangerous activities, such as construction, football, and motorcycling. One particular injury that a person may receive during such activities, even when wearing a helmet, is a concussion. Concussion injuries are very common, with between 1.7 and 3 million sports- and recreation-related concussions happening each year. Moreover, nearly half of all concussions go unreported or undetected. Three or more concussions can result in long-term cognitive problems, such as forgetfulness, foggy thinking, balance issues, or even trouble with eyesight. Wearing a helmet can reduce the severity of concussive injuries, but there remains a need for improved headgear that can protect the head and brain of a wearer, providing improved collision impact protection and reducing the number and severity of concussive injuries.

BRIEF SUMMARY OF THE INVENTION

[0008] An embodiment of the disclosure meets the needs presented above by generally comprising a helmet shell comprising a peripheral wall having an internal surface and an external surface. A liner is coupled to the internal surface of the peripheral wall of the helmet shell. The liner has an outer side and an inner side. The outer side is positioned adjacent to the internal surface of the peripheral wall. The inner side is configured to cover a head of a user when the user is wearing the helmet shell. The liner covers the internal surface of the peripheral wall of the helmet shell. The liner absorbs a force of an impact to the external surface of the peripheral wall of the helmet shell as the force of the impact travels from the external surface through the liner whereby the liner is configured to inhibit the force of the impact from injuring the user.

[0009] The liner may further comprise a first layer that is on the outer side. The first layer comprises a multi-density polymer cushioning material. A second layer is coupled to the first layer. The second layer comprises an air bladder having an interior space that is filled with a gas. A third layer is on the inner side and is coupled to the second layer. The third layer comprises a viscoelastic rubber material that is configured to absorb the force of the impact and disperse energy from the force of the impact across the third layer.

[0010] There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

[0011] The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

[0012] The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

[0013] FIG. 1 is an isometric view of a shock absorbing helmet liner device according to an embodiment of the disclosure.

[0014] FIG. 2 is a cross-sectional view of an embodiment of the disclosure.

[0015] FIG. 3 is a rear view of an embodiment of the disclosure.

[0016] FIG. 4 is a cross-sectional view of an embodiment of the disclosure.

[0017] FIG. 5 is an exploded view of an embodiment of the disclosure.

[0018] FIG. 6 is a cross-sectional view of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0019] With reference now to the drawings, and in particular to FIGS. 1 through 6 thereof, a new helmet embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral 10 will be described.

[0020] As best illustrated in FIGS. 1 through 6, the shock absorbing helmet liner device generally comprises a helmet shell 12 comprising a peripheral wall 14 having an internal surface 16 and an external surface 18. The peripheral wall 14 generally has a lower edge 20 defining an opening 22 into the helmet shell 12. The peripheral wall 14 generally has a front side 24 and a back side 26. The front side 24 is generally configured to frame a face of a user when the user is wearing the helmet shell 12.

[0021] A liner 28 is coupled to the internal surface 16 of the peripheral wall 14 of the helmet shell 12. The liner 28 may be removably coupled to the internal surface 16, or may be positionable within the helmet shell 12 adjacent to the internal surface 16. The liner 28 generally has an outer side 30 and an inner side 32. The outer side 30 is positioned adjacent to the internal surface 16 of the peripheral wall 14 of the helmet shell 12. The inner side 32 is configured to cover a head of the user when the user is wearing the helmet shell 12. The liner 28 may cover the internal surface 16 of the peripheral wall 14 of the helmet shell 12. The liner 28 absorbs a force of an impact to the external surface 18 of the peripheral wall 14 of the helmet shell 12 as the force of the impact travels from the external surface 18 through the liner 28 whereby the liner 28 is configured to inhibit the force of the impact from injuring the user.

[0022] The liner 28 may further comprise a first layer 34 that is on the outer side 30. The first layer 34 generally comprises a multi-density polymer cushioning material. For example, the multi-density polymer cushioning material may be a GDEFY® VERSOCLOUD® spring cell material. The multi-density polymer cushioning material can absorb energy during a collision, thereby reducing the energy that travels inwardly through the liner 28 toward the user.

[0023] A second layer 36 of the liner 28 is coupled to the first layer 34. The second layer 36 generally comprises an air bladder 38 having an interior space 40 that may be filled with a gas. Alternatively, the air bladder 38 may be fillable with the gas. The gas within the air bladder 38 has a pressure. The air bladder 38 may comprise a plastic material that is substantially nonporous, or that has very few pores, wherein the plastic material is configured to inhibit the gas from leaking out of the air bladder 38 thereby inhibiting the pressure of the gas from decreasing. The gas may comprise nitrogen, or another gas that has an atom size that is larger than a size of an oxygen atom

wherein the atom size is configured to inhibit the gas from leaking out of the air bladder **38**. In alternative embodiments, oxygen, air, or another appropriate gas may be used.

[0024] A third layer **42** of the liner **28** is coupled to the second layer **36**. The third layer **42** is on the inner side **32** of the liner **28**. The third layer **42** generally comprises a viscoelastic rubber material that is configured to absorb the force of the impact and disperse energy from the force of the impact across the third layer **42**. For example, the viscoelastic rubber material may be a NOENE® material.

[0025] A coupler **44** may releasably attach the outer side **30** of the liner **28** to the internal surface **16** of the peripheral wall **14** of the helmet shell **12**. The coupler **44** may comprise a first mating member **46** that is attached to the internal surface **16** of the peripheral wall **14** of the helmet shell **12**. A second mating member **48** may be releasably engageable with the first mating member **46**. The second mating member **48** is generally attached to the outer side **30** of the liner **28**. The second mating member **48** is alignable with the first mating member **46** when the liner **28** is positioned within the helmet shell **12**. For example, each of the first mating member **46** and the second mating member **48** may comprise a hook and loop material.

[0026] An air valve **50** may be coupled to the second layer **36** of the liner **28**. The air valve **50** is fluidically coupled to the interior space **40** of the air bladder **38** wherein the air valve **50** facilitates adding the gas to the interior space **40**. The air valve **50** may be positioned proximate to the lower edge **20** of the peripheral wall **14** wherein the air valve **50** is configured to be accessible to the user when the liner **28** is positioned within the helmet shell **12**.

[0027] For example, the air valve may include a tube **62** having an inlet **64** and an outlet **66**. The tube **62** may be positioned within an aperture **68** on the helmet shell **12**. The aperture **68** may extend into the external surface **18** of the helmet shell **12** and through the first layer **34** of the liner **28**, as shown in FIG. 4. The inlet **64** of the tube may be coupled to the external surface **18** and the outlet **66** of the tube may be coupled to the air bladder **38** wherein the tube **64** is fluidly coupled to the interior space **40** of the air bladder **38**. In such embodiments, the inlet **64** may be accessible from the back side **26** of the helmet shell **12** to add the gas to the interior space **40** of the air bladder **38**.

[0028] An alternative example is shown in FIG. 6. The tube **62** of the air valve **50** may extend downwardly from a terminal edge **70** of the air bladder **38**. The inlet **64** may be coupled to the internal surface **16** proximate to the lower edge **20** of the peripheral wall **14** of the helmet shell **12**. The outlet **66** is generally coupled to the air bladder **38** to fluidly couple the air valve **50** with the interior space **40** of the air bladder **38**.

[0029] A fastener **52** may releasably couple the air valve **50** to the internal surface **16** of the peripheral wall **14** of the helmet shell **12**. The fastener **52** may comprise a primary fastening portion **54** that is attached to the internal surface **16** of the peripheral wall **14** of the helmet shell **12**. A secondary fastening portion **56** may be releasably couplable with the primary fastening portion **54**. The secondary fastening portion **56** is generally attached to the air valve **50**. The secondary fastening portion **56** is alignable with the first fastening portion **54** when the liner **28** is positioned within the helmet shell **12**. For example, each of the primary fastening portion **54** and the secondary fastening portion **56** may comprise a hook and loop material.

[0030] A display panel **58** may be coupled to the peripheral wall **14** of the helmet shell **12**. The display panel **58** is electrically coupled to the second layer **36** of the liner **28**. The display panel **58** displays the pressure of the gas within the interior space **40** of the air bladder **38**. The display panel **58** may be positioned on the back side **26** of the peripheral wall **14**. The display panel **58** may be positioned adjacent to the lower edge **20** of the peripheral wall **14**.

[0031] A pressure sensor **60** may be electrically coupled to the display panel **58**. The pressure sensor **60** is coupled to the air bladder **38** wherein the pressure sensor **60** is configured to indicate the pressure of the gas within the interior space **40** of the air bladder **38**. The pressure sensor **60** is in communication with the display panel **58** wherein the display panel **58** is configured to display

the pressure indicated by the pressure sensor **60**.

[0032] In use, the liner **28** may be permanently attached to the helmet shell **12**, for example during manufacture of the helmet shell **12**. Alternatively, the liner **28** may be removably couplable to, or positionable within, the helmet shell **12**.

[0033] The multi-density polymer cushioning material of the first layer **34** can absorb energy during a collision, thereby reducing the energy that travels inwardly through the liner **28** toward the user. The second layer **36** can quickly distribute the gas within the air bladder **38**. Redistribution of the gas can redistribute the energy of the collision around the liner **28**. For example, the gas can redistribute spherically around the air bladder **38** after a point collision. This redistribution of the gas and the energy of the collision can reduce the unidirectional impact of a violent collision to the user. The air bladder **38** quickly converts unidirectional local impact forces into globally dispersed spherical forces to reduce rotational forces which could injure the user. When the air bladder **38** is locally compressed, the air bladder **38** quickly generates compressive forces in all spherical directions around the helmet shell **12**. The first layer **34** and the third layer **42** can absorb some of the energy redirected by the second layer **36**. The movement of the gas through the air bladder **38** can also create some push back, or recoil, on the opposite side of the helmet shell **12** from the impact to further reduce rotational forces and oscillation of the helmet. The second layer **36** thus better stabilizes the head and the brain of the user wearing the helmet. The air bladder **38** also provides compression within the liner **28** to better cushion the head of the user.

[0034] The pressure of the gas within the air bladder **38** is important for providing protection, so the pressure may be monitored regularly using the air valve **50** and the display panel **58**. In embodiments with the display panel **58**, the user can easily check the pressure of the gas within the air bladder **38** before every usage. Additionally, the large size of the atoms of the gas and the nonporous plastic material of the air bladder **38** can inhibit the gas from leaking out of the air bladder **38** and therefore inhibit reduction in the pressure of the gas over time.

[0035] The viscoelastic rubber material of the third layer **42** can convert energy from the impact to heat. The viscoelastic rubber material can absorb some of the energy and other forces of the impact. In particular, NOENE®, which is typically used for shoe insoles, can absorb up to 92% of the shock generated with each step taken by the person wearing the shoe insoles. The viscoelasticity of NOENE® makes it an excellent shock absorber, and it can absorb and then disperse energy from the impact across the third layer **42**.

[0036] Each of the layers of the liner **28** may be relatively thin. For example, the liner **28** may only reduce the size of the helmet shell **12** by between 0.25 inch and 0.5 inch, making the liner **28** useful for modifying a helmet that the user already owns or uses. Alternatively, new helmets could be manufactured with the liner **28** attached to the helmet shell **12**.

[0037] With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

[0038] Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. A protective headgear assembly comprising: a helmet shell comprising a peripheral wall having an internal surface and an external surface; a liner being coupled to the internal surface of the peripheral wall of the helmet shell, the liner having an outer side and an inner side, the outer side being positioned adjacent to the internal surface of the peripheral wall, the inner side being configured to cover a head of a user when the user is wearing the helmet shell wherein the liner absorbs a force of an impact to the external surface of the peripheral wall of the helmet shell whereby the liner is configured to inhibit the force of the impact from injuring the user, the liner further comprising: a first layer being on the outer side, the first layer comprising a multi-density polymer cushioning material; a second layer being coupled to the first layer, the second layer comprising an air bladder having an interior space being filled with a gas; and a third layer being coupled to the second layer, the third layer being on the inner side, the third layer comprising a viscoelastic rubber material being configured to absorb the force of the impact and disperse energy from the force of the impact across the third layer.
2. The protective headgear assembly of claim 1, wherein the multi-density polymer cushioning material comprises a GDEFY® VERSOCLOUD® spring cell material.
3. The protective headgear assembly of claim 1, wherein the gas has a pressure, the air bladder comprising a plastic material being nonporous wherein the plastic material is configured to inhibit the gas from leaking out of the air bladder thereby maintaining the pressure of the gas within the air bladder.
4. The protective headgear assembly of claim 3, wherein the peripheral wall has a lower edge defining an opening into the helmet shell, the peripheral wall having a back side, the protective headgear assembly further comprising a display panel being coupled to the peripheral wall of the helmet shell, the display panel being electrically coupled to the second layer of the liner, the display panel displaying the pressure of the gas within the interior space of the air bladder, the display panel being positioned on the back side of the peripheral wall, the display panel being positioned adjacent to the lower edge of the peripheral wall.
5. The protective headgear assembly of claim 1, wherein the gas comprises nitrogen.
6. The protective headgear assembly of claim 1, wherein the viscoelastic rubber material comprises a NOENE® material.
7. The protective headgear assembly of claim 1, further comprising a coupler releasably attaching the outer side of the liner to the internal surface of the peripheral wall of the helmet shell.
8. The protective headgear assembly of claim 7, the coupler further comprising: a first mating member being attached to the internal surface of the peripheral wall of the helmet shell; and a second mating member being releasably engageable with the first mating member, the second mating member being attached to the outer side of the liner, the second mating member being alignable with the first mating member when the liner is positioned within the helmet shell.
9. The protective headgear assembly of claim 8, wherein each of the first mating member and the second mating member comprise a hook and loop material.
10. The protective headgear assembly of claim 1, further comprising an air valve being coupled to the second layer of the liner, the air valve being fluidically coupled to the interior space of the air bladder wherein the air valve facilitates adding the gas to the interior space.
11. The protective headgear assembly of claim 10, wherein the peripheral wall has a lower edge defining an opening into the helmet shell, the air valve being positioned proximate to the lower edge of the peripheral wall wherein the air valve is configured to be accessible to the user when the liner is positioned within the helmet shell.
12. The protective headgear assembly of claim 10, further comprising a fastener releasably coupling the air valve to the internal surface of the peripheral wall of the helmet shell.

13. The protective headgear assembly of claim 12, the fastener comprising: a primary fastening portion being attached to the internal surface of the peripheral wall of the helmet shell; a secondary fastening portion being releasably couplable with the primary fastening portion, the secondary fastening portion being attached to the air valve, the secondary fastening portion being alignable with the first fastening portion when the liner is positioned within the helmet shell.

14. The protective headgear assembly of claim 13, wherein each of the primary fastening portion and the secondary fastening portion comprises a hook and loop material.

15. The protective headgear assembly of claim 1, further comprising a display panel being coupled to the peripheral wall of the helmet shell, the display panel being electrically coupled to the second layer of the liner, the display panel displaying a pressure of the gas within the interior space of the air bladder.

16. The protective headgear assembly of claim 1, wherein the liner is removably coupled to the internal surface of the peripheral wall of the helmet shell.

17. A protective headgear assembly comprising: a helmet shell comprising a peripheral wall having an internal surface and an external surface, the peripheral wall having a lower edge defining an opening into the helmet shell, the peripheral wall having a front side and a back side; a liner being coupled to the internal surface of the peripheral wall of the helmet shell, the liner having an outer side and an inner side, the outer side being positioned adjacent to the internal surface of the peripheral wall, the inner side being configured to cover a head of a user when the user is wearing the helmet shell, the liner covering the internal surface of the peripheral wall of the helmet shell wherein the liner absorbs a force of an impact to the external surface of the peripheral wall of the helmet shell as the force of the impact travels from the external surface through the liner whereby the liner is configured to inhibit the force of the impact from injuring the user, the liner further comprising: a first layer being on the outer side, the first layer comprising a multi-density polymer cushioning material, the multi-density polymer cushioning material being a GDEFY® VERSOCLOUD® spring cell material; a second layer being coupled to the first layer, the second layer comprising an air bladder having an interior space being filled with a gas, the gas having a pressure, the air bladder comprising a plastic material being nonporous wherein the plastic material is configured to inhibit the gas from leaking out of the air bladder, the gas comprising nitrogen; a third layer being coupled to the second layer, the third layer being on the inner side, the third layer comprising a viscoelastic rubber material being configured to absorb the force of the impact and disperse energy from the force of the impact across the third layer, the viscoelastic rubber material being NOENE® material; a coupler releasably attaching the outer side of the liner to the internal surface of the peripheral wall of the helmet shell, the coupler comprising: a first mating member being attached to the internal surface of the peripheral wall of the helmet shell; a second mating member being releasably engageable with the first mating member, the second mating member being attached to the outer side of the liner, the second mating member being alignable with the first mating member when the liner is positioned within the helmet shell; wherein each of the first mating member and the second mating member comprise a hook and loop material; an air valve being coupled to the second layer of the liner, the air valve being fluidically coupled to the interior space of the air bladder wherein the air valve facilitates adding the gas to the interior space, the air valve being positioned proximate to the lower edge of the peripheral wall wherein the air valve is configured to be accessible to the user when the liner is positioned within the helmet shell; a fastener releasably coupling the air valve to the internal surface of the peripheral wall of the helmet shell, the fastener comprising: a primary fastening portion being attached to the internal surface of the peripheral wall of the helmet shell; a secondary fastening portion being releasably couplable with the primary fastening portion, the secondary fastening portion being attached to the air valve, the secondary fastening portion being alignable with the first fastening portion when the liner is positioned within the helmet shell; wherein each of the primary fastening portion and the secondary fastening portion comprises a hook and loop material; and a display panel being coupled to the

peripheral wall of the helmet shell, the display panel being electrically coupled to the second layer of the liner, the display panel displaying the pressure of the gas within the interior space of the air bladder, the display panel being positioned on the back side of the peripheral wall, the display panel being positioned adjacent to the lower edge of the peripheral wall.
